

MESSINA, Italy

WMO: 164200

Lat: 38.2005N

Lon: 15.5530E

Elev: 51

StdP: 100.71

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.0	7.3	-3.4	2.9	9.6	-0.9	3.5	10.8	9.6	14.6	8.8	14.6	2.2	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	33.2	24.3	31.8	24.2	30.8	24.2	27.6	30.0	26.8	29.4	26.0	28.8	3.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.8	29.1	26.0	21.5	28.5	25.2	20.4	28.0	88.0	29.8	84.1	29.4	81.0	28.9	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.4	3.8	36.7	1.6	2.9	2.6	38.7	1.6	40.4	0.7	42.0	-0.5	44.1		
(5)			1.5	28.5	1.3	2.1	0.6	30.0	-0.2	31.2	-0.9	32.4	-1.9	33.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.3	12.5	12.3	14.1	16.4	20.1	24.4	27.5	27.8	24.4	21.0	17.1	13.9	(6)
(7)		DBStd	5.93	2.24	2.41	2.32	2.18	2.33	2.66	1.90	1.80	2.31	2.18	2.40	2.43	(7)
(8)		HDD10.0	18	6	7	2	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD18.3	749	182	168	133	66	8	1	0	0	0	4	49	139	(9)
(10)		CDD10.0	3422	83	72	128	191	313	432	542	553	433	340	213	123	(10)
(11)		CDD18.3	1112	0	0	1	6	62	182	283	294	183	86	13	1	(11)
(12)		CDH23.3	8746	0	0	3	8	165	1212	2869	3127	1174	179	7	1	(12)
(13)	CDH26.7	2628	0	0	1	1	18	301	971	1101	224	11	1	0	(13)	
(14)	Wind	WSAvg	2.8	2.8	3.0	3.0	3.3	3.1	2.9	2.9	2.6	2.4	2.5	2.7	2.7	(14)
(15)	Precipitation	PrecAvg	863	118	94	89	63	34	17	19	28	73	106	111	124	(15)
(16)		PrecMax	1527	308	247	233	168	220	79	83	137	180	371	253	324	(16)
(17)		PrecMin	464	17	12	1	4	0	0	0	0	1	5	26	14	(17)
(18)		PrecStd	175	64	49	55	40	33	20	21	32	45	72	51	58	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	19.8	22.1	24.2	28.8	34.7	35.7	35.1	32.0	27.9	24.2	21.4	(19)
MCWB			15.2	14.4	15.3	16.2	19.9	24.5	23.0	24.7	24.4	21.9	20.3	17.6	(20)	
2%		DB	17.2	17.6	19.9	22.3	26.7	31.1	33.1	33.2	30.1	26.2	22.5	19.1	(21)	
		MCWB	14.5	14.0	14.7	15.3	19.6	23.1	23.7	24.9	23.7	21.2	18.7	16.0	(22)	
5%		DB	16.3	16.4	18.5	20.9	25.1	29.7	31.7	31.9	28.9	25.1	21.4	18.1	(23)	
		MCWB	13.6	13.4	14.0	15.2	18.8	22.8	24.1	24.6	22.9	20.7	18.0	15.4	(24)	
10%		DB	15.6	15.6	17.3	19.8	23.8	28.2	30.7	30.8	27.9	24.2	20.5	17.2	(25)	
		MCWB	12.9	12.7	13.6	15.1	18.2	22.3	24.2	24.5	22.9	20.4	17.4	14.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	16.1	17.2	19.1	23.6	27.7	28.8	29.0	27.1	24.0	21.5	18.3	(27)
MCDWB			17.5	17.7	19.8	21.3	26.2	30.1	30.7	31.2	29.2	25.3	23.4	19.9	(28)	
2%		WB	15.1	15.0	16.1	17.7	21.9	26.0	27.4	27.8	25.9	22.8	19.9	17.1	(29)	
		MCDWB	16.5	16.7	18.1	19.9	24.4	29.1	29.8	30.2	28.1	24.8	21.4	18.5	(30)	
5%		WB	14.3	14.1	15.2	16.8	20.4	24.9	26.6	26.9	24.9	21.9	18.9	16.0	(31)	
		MCDWB	15.9	15.7	17.2	19.1	23.2	28.0	29.3	29.6	27.2	24.0	20.8	17.5	(32)	
10%		WB	13.4	13.3	14.5	16.1	19.4	23.7	25.9	26.2	24.0	21.1	18.0	14.9	(33)	
		MCDWB	15.2	15.1	16.5	18.4	22.3	26.9	28.7	29.0	26.6	23.4	19.9	16.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	4.7	5.1	5.5	5.8	6.0	6.2	6.0	5.6	5.2	4.8	4.3	(35)
MCDBR			4.6	5.2	6.2	7.0	7.2	7.4	7.3	7.1	6.4	5.9	5.4	5.0	(36)	
MCWBR			3.4	3.5	4.0	4.5	4.7	5.1	5.3	5.4	4.2	3.5	3.8	3.7	(37)	
5% WB		MCDWB	4.3	4.8	5.9	6.2	6.6	6.9	6.7	6.5	5.8	5.4	5.0	4.7	(38)	
		MCWBR	3.7	3.8	4.3	4.6	4.9	5.3	5.3	5.6	4.7	4.0	4.3	4.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.350	0.371	0.412	0.445	0.445	0.428	0.426	0.432	0.436	0.414	0.380	0.355	(40)	
taud		2.454	2.373	2.252	2.176	2.228	2.316	2.340	2.322	2.311	2.333	2.396	2.443	(41)		
Ebn at Noon		823	849	845	836	842	854	852	839	812	789	776	786	(42)		
Edh at Noon		84	102	127	145	140	128	124	124	119	105	88	80	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.13	2.92	4.23	5.44	6.55	7.29	7.46	6.59	4.87	3.52	2.39	1.89	(44)	
(45)		RadStd	0.17	0.25	0.32	0.36	0.35	0.38	0.23	0.33	0.23	0.27	0.18	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.35	N/A	N/A	N/A	N/A

Nomenclature: See separate page